Further Reading: [Kay98] provides detailed, readable coverage of hypothesis testing. [Hay01] presents detection of digital communications signals as a hypothesis test. A collection of challenging homework problems for sections 11.3 and 11.4 are based on bit detection for code division multiple access (CDMA) communications systems. The authoritative treatment of this subject can be found in [Ver98].

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

11.1.1● Let L equal the number of flips of a coin up to and including the first flip of heads. Devise a significance test for L at level $\alpha = 0.05$ to test the hypothesis H that the coin is fair. What are the limitations of the test?

11.1.2● A course has two recitation sections that meet at different times. On the midterm, the average for section 1 is 5 points higher than the average for section 2. A logical conclusion is that the TA for section 1 is better than the TA for section 2. Using words rather than math, give reasons why this might be the wrong conclusion.

11.1.3● Under the null hypothesis H_0 that traffic is typical, the number of call attempts in a 1-second interval (during rush hour) at a mobile telephone switch is a Poisson random variable N with $\mathrm{E}[N] = 2.5$. Over a T-second period, the measured call rate is $M = (N_1 + \cdots + N_T)/T$, where $N_1, \ldots, N_T$ are iid Poisson random variables identical to N. However, whenever there is unusually heavy traffic (resulting from an accident or bad weather or some other event), the measured call rate M is higher than usual. Based on the observation M, design a significance test to reject the null hypothesis H_0 that traffic is typical at a significance level $\alpha = 0.05$. Justify your choice of the rejection region R. Hint: You may use a Gaussian (central limit theorem) approximation for calculating probabilities with respect to M. How does your test depend on the observation period T? Explain your answer.

11.1.4● A cellular telephone company is upgrading its network to a new (N) transmission system one area at a time, but they do not announce where the upgrades take place. You have the task of determining whether certain areas have been upgraded. You have decided to use an application in your smartphone to measure the ping time (how long it takes to receive a response to a certain message) in each area. The new system is faster than the old (O) one. It has on average shorter ping times. The probability model for the ping time in milliseconds of the new system is the exponential (60) random variable. Perform a ping test and reject the null hypothesis that the area has the new system if the ping time is greater than t_0 ms.

(a) Write a formula for α, the significance of the test as a function of t_0.

(b) What is the value of t_0 that produces a significance level $\alpha = 0.05$?

11.1.5● When a pacemaker factory is operating normally (the null hypothesis H_0), a randomly selected pacemaker fails a "drop test" with probability $q_0 = 10^{-4}$. Each day, an inspector randomly tests pacemakers. Design a significance test for the null hypothesis with significance level $\alpha = 0.01$. Note that drop testing of pacemakers is expensive because the pacemakers that are tested must be discarded. Thus the significance test should try to minimize the number of pacemakers tested.

11.1.6■ Let K be the number of heads in $n = 100$ flips of a coin. Devise significance tests for the hypothesis H that the coin is fair such that

(a) The significance level $\alpha = 0.05$ and the rejection set R has the form $\{|K - \mathrm{E}[K]| > c\}$.

(b) The significance level $\alpha = 0.01$ and the rejection set R has the form $\{K > c'\}$.

11.1.7■ When a chip fabrication facility is operating normally, the lifetime of a microchip operated at temperature T, measured in degrees Celsius, is given by an exponential (λ) random variable X with expected value $\mathrm{E}[X] = 1/\lambda = (200/T)^2$ years. Occasionally, the chip fabrication plant has contamination problems and the chips tend to fail much more rapidly. To test for contamination problems, each day m chips are subjected to a one-day test at $T = 100°C$. Based on the number N of chips that fail in one day, design a significance test for the null hypothesis test H_0 that the plant is operating normally.

(a) Suppose the rejection set of the test is $R = \{N > 0\}$. Find the significance level of the test as a function of m, the number of chips tested.

(b) How many chips must be tested so that the significance level is $\alpha = 0.01$.

(c) If we raise the temperature of the test, does the number of chips we need to test increase or decrease?

11.1.8■ A group of n people form a football pool. The rules of this pool are simple: 16 football games are played each week. Each contestant must pick the winner of each game against a point *spread*. The contestant who picks the most games correctly over a 16-week season wins the pool. The spread is a point difference d such that picking the favored team is a winning pick only if that team wins by more than d points; otherwise, the pick of the opposing team is a winner. Each pool contestant can study the teams' past histories, performance trends, official injury reports, coaches' weekly press conferences, chat room gossip and any other wisdom that might help in placing a winning bet.

After m weeks, contestant i will have picked W_i games correctly out of $16m$ games. For example, suppose that after $m = 14$ weeks, $16(14) = 224$ games have been played and that the leader (call him Narayan) has picked 119 games correctly. Does the pool leader Narayan have skills or is he just lucky?

(a) To address this question, design a significance test to determine whether the pool leader actually has any skill at picking games. Let H_0 denote the null hypothesis that *all* players, including the leader, pick winners in each game with probability $p = 1/2$, independent of the outcome of any other game. Based on the observation of W, the number of winning picks by the pool leader after m weeks of the season, design a one-sided significance test for hypothesis H_0 at significance level $\alpha = 0.05$. You may use a central limit theorem approximation for binomial PMFs as needed.

(b) Given that Narayan is the leader with 119 winning picks in $m = 14$ weeks in a pool with $n = 38$ contestants, do you reject or accept hypothesis H_0?

(c) How does the significance test depend on picks being made against the point spread?

11.1.9♦ A class has $2n$ (a large number) students The students are separated into two groups A and B, each with n students. Group A students take exam A and earn iid scores $X_1, \ldots, X_n$. Group B students take exam B, earning iid scores $Y_1, \ldots, Y_n$. The two exams are similar but different; however, the exams were designed so that a student's score X on exam A or Y on exam B have the same expected value and variance $\sigma^2 = 100$. For each exam, we form the sample mean statistic

$$M_A = \frac{X_1 + \cdots + X_n}{n},$$
$$M_B = \frac{Y_1 + \cdots + Y_n}{n}.$$

Based on the statistic $D = M_A - M_B$, use the central limit theorem to design a significance test at significance level $\alpha = 0.05$ for the hypothesis H_0 that a student's score on the two exams has the same expected value μ and variance $\sigma^2 = 100$. What is the rejection region if $n = 100$? Make sure to specify any additional assumptions that you need to make; however, try to make as few additional assumptions as possible.

11.2.1● In a random hour, the number of call attempts N at a telephone switch has a Poisson distribution with an expected value of either α_0 (hypothesis H_0) or α_1 (hypothesis H_1). For a priori probabilities $\mathrm{P}[H_i]$, find the MAP and ML hypothesis testing rules given the observation of N.

11.2.2■ The ping time, in milliseconds of a new transmission system, described in Problem 11.1.4 is the exponential (60) random variable N. The ping time of an old system is an exponential random variable O with expected value $\mu_O > 60$ ms. The null hypothesis of a binary hypothesis test is H_0: The transmission system is the new system. The alternative hypothesis is H_1: The transmission system is the old system. The probability of a new system is $\mathrm{P}[N] = 0.8$. The probability of an old system is $\mathrm{P}[O] = 0.2$. A binary hypothesis test measures T milliseconds, the result of one ping test. The decision is H_0 if $T \leq t_0$ ms. Otherwise, the decision is H_1.

(a) Write a formula for the false alarm probability as a function of t_0 and μ_O.

(b) Write a formula for the miss probability as a function of t_0 and μ_O.

(c) Calculate the maximum likelihood decision time $t_0 = t_{ML}$ for $\mu_O = 120$ ms and $\mu_O = 200$ ms.

(d) Do you think that t_{MAP}, the maximum a posteriori decision time, is greater than or less than t_{ML}? Explain your answer.

(e) Calculate the maximum a posteriori probability decision time $t_0 = t_{MAP}$ for $\mu_O = 120$ ms and $\mu_O = 200$ ms.

(f) Draw the receiver operating curves for $\mu_O = 120$ ms and $\mu_O = 200$ ms.

11.2.3■ An automatic doorbell system rings a bell whenever it detects someone at the door. The system uses a photodetector such that if a person is present, hypothesis H_1, the photodetector output N is a Poisson random variable with an expected value of 1300 photons. Otherwise, if no one is there, hypothesis H_0, the photodetector output is a Poisson random variable with an expected value of 1000. Devise a Neyman–Pearson test for the presence of someone outside the door such that the false alarm probability is $\alpha \leq 10^{-6}$. What is minimum value of P_{MISS}?

11.2.4■ In the radar system of Example 11.4, $\mathrm{P}[H_1] = 0.01$. In the case of a false alarm, the system issues an unnecessary alert at the cost of $C_{10} = 1$ unit. The cost of a miss is $C_{01} = 10^4$ units because the target could cause a lot of damage. When the target is present, the voltage is $X = 4+N$, a Gaussian $(4, 1)$ random variable. When there is no target present, the voltage is $X = N$, the Gaussian $(0, 1)$ random variable. In a binary hypothesis test, the acceptance sets are $A_0 = \{X \leq x_0\}$ and $A_1 = \{X > x_0\}$.

(a) For the MAP hypothesis test, find the decision threshold $x_0 = x_{\mathrm{MAP}}$, the error probabilities P_{FA} and P_{MISS}, and the average cost $\mathrm{E}[C]$.

(b) Compare the MAP test performance against the minimum cost hypothesis test.

11.2.5■ In the radar system of Example 11.4, show that the ROC in Figure 11.2 is the result of a Neyman–Pearson test. That is, show that the Neyman–Pearson test is a threshold test with acceptance set $A_0 = \{X \leq x_0\}$. How is x_0 related to the false alarm probability α?

11.2.6◆ A system administrator (and part-time spy) at a classified research facility wishes to use a gateway router for covert communication of research secrets to an outside accomplice. The sysadmin covertly communicates a bit W for every n transmitted packets. To signal $W = 0$, the router does nothing while n regular packets are sent out through the gateway as a

Poisson process of rate λ_0 packets/sec. To signal $W = 1$ the sysadmin injects additional fake outbound packets so that n outbound packets are sent as a Poisson process of rate $2\lambda_0$. The secret communication bits are equiprobable in that $\mathrm{P}[W = 1] = \mathrm{P}[W = 0] = 1/2$. The sysadmin's accomplice (outside the gateway) monitors the outbound packet transmission process by observing the vector $\mathbf{X} = [X_1, X_2, \ldots, X_n]$ of packet interarrival times and guessing the bit W every n packets.

(a) Find the conditional PDFs $f_{\mathbf{X}|W=0}(\mathbf{x})$ and $f_{\mathbf{X}|W=1}(\mathbf{x})$.

(b) What are the MAP and ML hypothesis tests for the accomplice to guess either hypothesis H_0 that $W = 0$ or hypothesis H_1 that $W = 1$?

(c) Let $\hat{W}$ denote the decision of the ML hypothesis test. Use the Chernoff bound to upper bound the error probability $\mathrm{P}[\hat{W} = 0|W = 1]$.

11.2.7♦ The ping time, in milliseconds, of a new transmission system, described in Problem 11.1.4 is the exponential (60) random variable N. The ping time of an old system is the exponential (120) random variable O. The null hypothesis of a binary hypothesis test is H_0: The transmission system is the new system. The alternative hypothesis is H_1: The transmission system is the old system. The probability of a new system is $\mathrm{P}[N] = 0.8$. The probability of an old system $\mathrm{P}[O] = 0.2$. A binary hypothesis test performs k ping tests and calculates $M_n(T)$, the sample mean of the ping time. The decision is H_0 if $M_n(T) \leq t_0$ ms. Otherwise, the decision is H_1.

(a) Use the central limit theorem to write a formula for the false alarm probability as a function of t_0 and k.

(b) Use the central limit theorem to write a formula for the miss probability as a function of t_0 and k.

(c) Calculate the maximum likelihood decision time, $t_0 = t_{\mathrm{ML}}$, for $k = 9$ ping tests.

(d) Calculate the maximum a posteriori probability decision time, $t_0 = t_{\mathrm{MAP}}$ for $k = 9$ ping tests.

(e) Draw the receiver operating curves for $k = 9$ ping tests and $k = 16$ ping tests.

11.2.8♦ In this problem, we perform the old/new detection test of Problem 11.2.7, except now we monitor k ping tests and observe whether each ping lasts longer than t_0 ms. The random variable M is the number of pings that last longer than t_0 ms. The decision is H_0 if $M \leq m_0$. Otherwise, the decision is H_1.

(a) Write a formula for the false alarm probability as a function of t_0, m_0, and n.

(b) Find the maximum likelihood decision number $m_0 = m_{ML}$ for $t_0 = 4.5$ ms and $k = 16$ ping tests.

(c) Find the maximum a posteriori probability decision number $m_0 = m_{MAP}$ for $t_0 = 4.5$ ms and $k = 16$ ping tests.

(d) Draw the receiver operating curves for $t_0 = 90$ ms and $t_0 = 60$ ms. In both cases let $k = 16$ ping tests.

11.2.9♦ A binary communication system has transmitted signal X, the Bernoulli (1/2) random variable. At the receiver, we observe $Y = VX + W$, where V is a "fading factor" and W is additive noise. Note that V and W are exponential (1) random variables and that X, V, and W are mutually independent. Given the observation Y, we must guess whether $X = 0$ or $X = 1$ was transmitted. Use a binary hypothesis test to determine the rule that minimizes the probability P_e of a decoding error. For the optimum decision rule, calculate P_e.

11.2.10♦ In a BPSK amplify-and-forward relay system, a source transmits a random bit $V \in \{-1, 1\}$ every T seconds to a destination receiver via a set of n relay transmitters. $V = 1$ and $V = -1$ are equally likely. In this communication system, the source transmits during the time period $(0, T/2)$

such that relay i receives

$$X_i = \alpha_i V + W_i, \qquad i = 1, 2, \ldots, n,$$

where the W_i are iid Gaussian $(0, 1)$ random variables representing relay i receiver noise. In the time interval $(T/2, T)$, each relay node amplifies and forwards the received source signal. The destination receiver obtains the vector $\mathbf{Y} = \begin{bmatrix} Y_1 & \cdots & Y_n \end{bmatrix}'$ such that

$$Y_i = \beta_i X_i + Z_i, \qquad i = 1, 2, \ldots, n,$$

where the Z_i are also iid Gaussian $(0, 1)$ random variables. In the following, assume that the parameters α_i and β_i are all nonnegative. Also, let H_0 denote the hypothesis that $V = -1$ and H_1 the hypothesis $V = 1$.

(a) Suppose you build a suboptimal detector based on the sum $Y = \sum_{i=1}^{n} Y_i$. If $Y > 0$, the receiver guesses H_1; otherwise the receiver guesses H_0. What is the probability of error P_e for this receiver?

(b) Based on the observation $\mathbf{Y}$, now suppose the destination receiver detector performs a MAP test for hypotheses H_0 or H_1. What is the MAP detector rule? Simplify your answer as much as possible. Hint: First find the likelihood functions $f_{\mathbf{Y}|H_i}(y)$.

(c) What is the probability of bit error P_e^* of the MAP detector?

(d) Compare the two detectors when $n = 4$ and

$$(\alpha_1, \beta_1) = (1, 1), \qquad (\alpha_2, \beta_2) = (10, 1),$$
$$(\alpha_3, \beta_3) = (1, 10), \qquad (\alpha_4, \beta_4) = (10, 10).$$

In general, what's bad about the suboptimal detector?

11.2.11◆ In a BPSK communication system, a source wishes to communicate a random bit $X \in \{-1, 1\}$ to a receiver. Inputs $X = 1$ and $X = -1$ are equally likely. In this system, the source transmits X multiple times. In the ith transmission, the receiver observes $Y_i = X + W_i$, where the W_i are iid Gaussian $(0, 1)$ noises, independent of X.

(a) After n transmissions of X, you observe $\mathbf{Y} = \mathbf{y} = \begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix}'$. Find $\mathrm{P}[X = 1|\mathbf{Y} = \mathbf{y}]$. Express your answer in terms of the likelihood ratio

$$L(\mathbf{y}) = \frac{f_{\mathbf{Y}|X}(\mathbf{y}|-1)}{f_{\mathbf{Y}|X}(\mathbf{y}|1)}.$$

(b) Suppose after n transmissions, the receiver observes $\mathbf{Y} = \mathbf{y}$ and decides

$$X^* = \begin{cases} 1 & \mathrm{P}\left[X = 1|\mathbf{Y} = \mathbf{y}\right] > 1/2, \\ -1 & \text{otherwise.} \end{cases}$$

Find the probability of error $P_e = \mathrm{P}[X^* \neq X]$ in terms of the $\Phi(\cdot)$ function. Hint: For P_e calculations, symmetry implies $P_e = \mathrm{P}[X^* \neq X|X = 1]$.

(c) Now suppose the system uses ARQ as follows. If $|\hat{X}_n(\mathbf{y})| < 1 - \epsilon$, the receiver requests that X be retransmitted; otherwise the receiver guesses what is transmitted. In particular, if $\hat{X}_n(\mathbf{y}) > 1 - \epsilon$, the receiver guesses $X^* = 1$. If $\hat{X}_n(\mathbf{y}) < -1 + \epsilon$, the receiver guesses $X^* = -1$. Following the receiver's guess, the transmitter starts sending a new bit. Find upper and lower bounds to $P_e = \mathrm{P}[X^* \neq X]$. That is, find ϵ_1 and ϵ_2 such that

$$\epsilon_1 \leq P_e \leq \epsilon_2.$$

11.2.12◆ Suppose in the disk drive factory of Example 11.8, we can observe K, the number of failed devices out of n devices tested. As in the example, let H_i denote the hypothesis that the failure rate is q_i.

(a) Assuming $q_0 < q_1$, what is the ML hypothesis test based on an observation of K?

(b) What are the conditional probabilities of error $P_{\text{FA}} = \mathrm{P}[A_1|H_0]$ and $P_{\text{MISS}} = \mathrm{P}[A_0|H_1]$? Calculate these probabilities for $n = 500, q_0 = 10^{-4}, q_1 = 10^{-2}$.

(c) Compare this test to that considered in Example 11.8. Which test is more reliable? Which test is easier to implement?

11.2.13♦ Consider a binary hypothesis test in which there is a cost associated with each type of decision. In addition to the cost C'_{10} for a false alarm and C'_{01} for a miss, we also have the costs C'_{00} for correctly deciding hypothesis H_0 and the C'_{11} for correctly deciding hypothesis H_1. Based on the observation of a continuous random vector $\mathbf{X}$, design the hypothesis test that minimizes the total expected cost

$$\begin{aligned}\mathrm{E}\left[C'\right] = &\mathrm{P}\left[A_1|H_0\right]\mathrm{P}\left[H_0\right]C'_{10}\\ &+\mathrm{P}\left[A_0|H_0\right]\mathrm{P}\left[H_0\right]C'_{00}\\ &+\mathrm{P}\left[A_0|H_1\right]\mathrm{P}\left[H_1\right]C'_{01}\\ &+\mathrm{P}\left[A_1|H_1\right]\mathrm{P}\left[H_1\right]C'_{11}.\end{aligned}$$

Show that the decision rule that minimizes total cost is the same as decision rule of the minimum cost test in Theorem 11.3, with the costs C_{01} and C_{10} replaced by the differential costs $C'_{01} - C'_{11}$ and $C'_{10} - C'_{00}$.

11.3.1● In a ternary amplitude shift keying (ASK) communications system, there are three equally likely transmitted signals $\{s_0, s_1, s_2\}$. These signals are distinguished by their amplitudes such that if signal s_i is transmitted, the receiver output will be

$$X = a(i-1) + N,$$

where a is a positive constant and N is a Gaussian $(0, \sigma_N)$ random variable. Based on the output X, the receiver must decode which symbol s_i was transmitted.

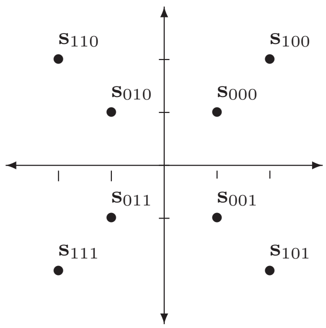
(a) What are the acceptance sets A_i for the hypotheses H_i that s_i was transmitted?

(b) What is $\mathrm{P}[D_E]$, the probability that the receiver decodes the wrong symbol?

11.3.2● A multilevel QPSK communications system transmits three bits every unit of time. For each possible sequence ijk of three bits, one of eight symbols, $\{s_{000}, s_{001}, \ldots, s_{111}\}$, is transmitted. When signal s_{ijk} is transmitted, the receiver output is

$$\mathbf{X} = \mathbf{s}_{ijk} + \mathbf{N},$$

where $\mathbf{N}$ is a Gaussian $(\mathbf{0}, \sigma^2\mathbf{I})$ random vector. The two-dimensional signal vectors $\mathbf{s}_{000}, \ldots, \mathbf{s}_{111}$ are



Let H_{ijk} denote the hypothesis that s_{ijk} was transmitted. The receiver output $\mathbf{X} = \begin{bmatrix}X_1 & X_2\end{bmatrix}'$ is used to decide the acceptance sets $\{A_{000}, \ldots, A_{111}\}$. If all eight symbols are equally likely, sketch the acceptance sets.

11.3.3■ An M-ary quadrature amplitude modulation (QAM) communications system can be viewed as a generalization of the QPSK system described in Example 11.13. In the QAM system, one of M equally likely symbols $s_0, \ldots, s_{m-1}$ is transmitted every unit of time. When symbol s_i is transmitted, the receiver produces the two-dimensional vector output

$$\mathbf{X} = \mathbf{s}_i + \mathbf{N},$$

where $\mathbf{N}$ has iid Gaussian $(0, \sigma^2)$ components. Based on the output $\mathbf{X}$, the receiver must decide which symbol was transmitted. Design a hypothesis test that maximizes the probability of correctly deciding what symbol was sent. Hint: Following Example 11.13, describe the acceptance set in terms of the vectors

$$\mathbf{x} = \begin{bmatrix}x_1\\x_2\end{bmatrix}, \qquad \mathbf{s}_i = \begin{bmatrix}s_{i1}\\s_{i2}\end{bmatrix}.$$

11.3.4 Suppose a user of the multilevel QPSK system needs to decode only the third bit k of the message ijk. For $k = 0, 1$, let H_k denote the hypothesis that the third bit was k. What are the acceptance sets A_0 and A_1? What is $\mathrm{P}[B_3]$, the probability that the third bit is in error?

11.3.5 The QPSK system of Example 11.13 can be generalized to an M-ary phase shift keying (MPSK) system with $M > 4$ equally likely signals. The signal vectors are $\{\mathbf{s}_0, \ldots, \mathbf{s}_{M-1}\}$, where

$$\mathbf{s}_i = \begin{bmatrix} s_{i1} \\ s_{i2} \end{bmatrix} = \begin{bmatrix} \sqrt{E}\cos\theta_i \\ \sqrt{E}\sin\theta_i \end{bmatrix}$$

and $\theta_i = 2\pi i/M$. When the ith message is sent, the received signal is $\mathbf{X} = \mathbf{s}_i + \mathbf{N}$ where $\mathbf{N}$ is a Gaussian $(\mathbf{0}, \sigma^2\mathbf{I})$ noise vector.

(a) Sketch the acceptance set A_i for the hypothesis H_i that $\mathbf{s}_i$ was transmitted.

(b) Find the largest value of d such that

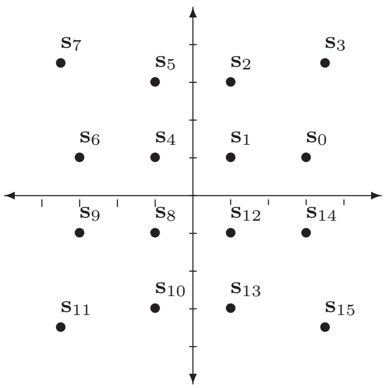
$$\{\mathbf{x} |\, \|\mathbf{x} - \mathbf{s}_i\| \le d\} \subset A_i.$$

(c) Use d to find an upper bound for the probability of error.

11.3.6 A modem uses QAM (see Problem 11.3.3) to transmit one of 16 symbols, $s_0, \ldots, s_{15}$, every 1/600 seconds. When signal s_i is transmitted, the receiver output is

$$\mathbf{X} = \mathbf{s}_i + \mathbf{N}.$$

The signal vectors $\mathbf{s}_0, \ldots, \mathbf{s}_{15}$ are



(a) Sketch the acceptance sets based on the receiver outputs X_1, X_2. Hint: Apply the solution to Problem 11.3.3.

(b) Let H_i be the event that symbol s_i was transmitted and let C be the event that the correct symbol is decoded. What is $\mathrm{P}[C|H_1]$?

(c) Argue that $\mathrm{P}[C] \ge \mathrm{P}[C|H_1]$.

11.3.7 For the QPSK communications system of Example 11.13, identify the acceptance sets for the MAP hypothesis test when the symbols are not equally likely. Sketch the acceptance sets when $\sigma = 0.8$, $E = 1$, $\mathrm{P}[H_0] = 1/2$, and $\mathrm{P}[H_1] = \mathrm{P}[H_2] = \mathrm{P}[H_3] = 1/6$.

11.3.8 In a code division multiple access (CDMA) communications system, k users share a radio channel using a set of n-dimensional code vectors $\{\mathbf{S}_1, \ldots, \mathbf{S}_k\}$ to distinguish their signals. The dimensionality factor n is known as the processing gain. Each user i transmits independent data bits X_i such that the vector $\mathbf{X} = \begin{bmatrix} X_1 & \cdots & X_k \end{bmatrix}'$ has iid components with $P_{X_i}(1) = P_{X_i}(-1) = 1/2$. The received signal is

$$\mathbf{Y} = \sum_{i=1}^{k} X_i \sqrt{p_i}\mathbf{S}_i + \mathbf{N},$$

where $\mathbf{N}$ is a Gaussian $(\mathbf{0}, \sigma^2\mathbf{I})$ noise vector. From the observation $\mathbf{Y}$, the receiver performs a multiple hypothesis test to decode the data bit vector $\mathbf{X}$.

(a) Show that in terms of vectors,

$$\mathbf{Y} = \mathbf{S}\mathbf{P}^{1/2}\mathbf{X} + \mathbf{N},$$

where $\mathbf{S}$ is an $n \times k$ matrix with ith column $\mathbf{S}_i$ and $\mathbf{P}^{1/2} = \mathrm{diag}[\sqrt{p_1}, \ldots, \sqrt{p_k}]$ is a $k \times k$ diagonal matrix.

(b) Given $\mathbf{Y} = \mathbf{y}$, show that the MAP and ML detectors for $\mathbf{X}$ are the same and are given by

$$\mathbf{x}^*(\mathbf{y}) = \arg\min_{\mathbf{x} \in B_k} \left\| \mathbf{y} - \mathbf{S}\mathbf{P}^{1/2}\mathbf{x} \right\|.$$

where B_k is the set of all k dimensional vectors with ± 1 elements.

(c) How many hypotheses does the ML detector need to evaluate?

11.3.9◆ For the CDMA communications system of Problem 11.3.8, a detection strategy known as *decorrelation* applies a transformation to $\mathbf{Y}$ to generate

$$\tilde{\mathbf{Y}} = (\mathbf{S}'\mathbf{S})^{-1}\mathbf{S}'\mathbf{Y} = \mathbf{P}^{1/2}\mathbf{X} + \tilde{\mathbf{N}}$$

where $\tilde{\mathbf{N}} = (\mathbf{S}'\mathbf{S})^{-1}\mathbf{S}'\mathbf{N}$ is still a Gaussian noise vector with expected value $\mathrm{E}[\tilde{\mathbf{N}}] = \mathbf{0}$. Decorrelation separates the signals in that the ith component of $\tilde{\mathbf{Y}}$ is

$$\tilde{Y}_i = \sqrt{p_i} X_i + \tilde{N}_i,$$

which is the same as a single-user receiver output of the binary communication system of Example 11.6. For equally likely inputs $X_i = 1$ and $X_i = -1$, Example 11.6 showed that the optimal (minimum probability of bit error) decision rule based on the receiver output $\tilde{Y}_i$ is

$$\hat{X}_i = \operatorname{sgn}(\tilde{Y}_i).$$

Although this technique requires the code vectors $\mathbf{S}_1, \ldots, \mathbf{S}_k$ to be linearly independent, the number of hypotheses that must be tested is greatly reduced in comparison to the optimal ML detector introduced in Problem 11.3.8. In the case of linearly independent code vectors, is the decorrelator optimal? That is, does it achieve the same bit error rate (BER) as the optimal ML detector?

11.4.1● A wireless pressure sensor (buried in the ground) reports a discrete random variable X with range $S_X = \{1, 2, \ldots, 20\}$ to signal the presence of an object. Given an observation X and a threshold x_0, we decide that an object is present (hypothesis H_1) if $X > x_0$; otherwise we decide that no object is present (hypothesis H_0). Under hypothesis H_i, X has conditional PMF

$$P_{X|H_i}(x) = \begin{cases} \frac{(1-p_i)p_i^{x-1}}{1-p_i^{20}} & x = 1, 2, \ldots, 20, \\ 0 & \text{otherwise}, \end{cases}$$

where $p_0 = 0.99$ and $p_1 = 0.9$. Calculate and plot the false alarm and miss probabilities as a function of the detection threshold x_0. Calculate the discrete receiver operating curve (ROC) specified by x_0.

11.4.2● For the binary communications system of Example 11.7, graph the error probability P_{ERR} as a function of p, the probability that the transmitted signal is 0. For the signal-to-noise voltage ratio, consider $v/\sigma \in \{0.1, 1, 10\}$. What values of p minimize P_{ERR}? Why are those values not practical?

11.4.3● For the squared distortion communications system of Example 11.14 with $v = 1.5$ and $d = 0.5$, find the value of T that minimizes P_e.

11.4.4■ A poisonous gas sensor reports continuous random variable X. In the presence of toxic gases, hypothesis H_1,

$$f_{X|H_1}(x) = \begin{cases} (x/8)e^{-x^2/16} & x \geq 0, \\ 0 & \text{otherwise}. \end{cases}$$

In the absence of dangerous gases, X has conditional PDF

$$f_{X|H_0}(x) = \begin{cases} (1/2)e^{-x/2} & x \geq 0, \\ 0 & \text{otherwise}. \end{cases}$$

Devise a hypothesis test that determines the presence of poisonous gases. Plot the false alarm and miss probabilities for the test as a function of the decision threshold. Lastly, plot the corresponding receiver operating curve.

11.4.5◆ Simulate the M-ary PSK system in Problem 11.3.5 for $M = 8$ and $M = 16$. Let $\hat{P}_{\mathrm{ERR}}$ denote the relative frequency of symbol errors in the simulated transmission in 10^5 symbols. For each value of M, graph $\hat{P}_{\mathrm{ERR}}$, as a function of the signal-to-noise power ratio (SNR) $\gamma = E/\sigma^2$. Consider

$10\log_{10}\gamma$, the SNR in dB, ranging from 0 to 30 dB.

11.4.6◆◆ In this problem, we evaluate the bit error rate (BER) performance of the CDMA communications system introduced in Problem 11.3.8. In our experiments, we will make the following additional assumptions.

- In practical systems, code vectors are generated pseudorandomly. We will assume the code vectors are random. For each transmitted data vector $\mathbf{X}$, the code vector of user i will be $\mathbf{S}_i = \frac{1}{\sqrt{n}}\begin{bmatrix} S_{i1} & S_{i2} & \cdots & S_{in}\end{bmatrix}'$, where the components S_{ij} are iid random variables such that $P_{S_{ij}}(1) = P_{S_{ij}}(-1) = 1/2$. Note that the factor $1/\sqrt{n}$ is used so that each code vector $\mathbf{S}_i$ has length 1: $\|\mathbf{S}_i\|^2 = \mathbf{S}_i'\mathbf{S}_i = 1$.
- Each user transmits at 6dB SNR. For convenience, assume $P_i = p = 4$ and $\sigma^2 = 1$.

(a) Use MATLAB to simulate a CDMA system with processing gain $n = 16$. For each experimental trial, generate a random set of code vectors $\{\mathbf{S}_i\}$, data vector $\mathbf{X}$, and noise vector $\mathbf{N}$. Find the ML estimate $\mathbf{x}^*$ and count the number of bit errors; i.e., the number of positions in which $x_i^* \neq X_i$. Use the relative frequency of bit errors as an estimate of the probability of bit error. Consider $k = 2, 4, 8, 16$ users. For each value of k, perform enough trials so that bit errors are generated on 100 independent trials. Explain why your simulations take so long.

(b) For a simpler detector known as the matched filter, when $\mathbf{Y} = \mathbf{y}$, the detector decision for user i is

$$\hat{x}_i = \operatorname{sgn}(\mathbf{S}_i'\mathbf{y})$$

where $\operatorname{sgn}(x) = 1$ if $x > 0$, $\operatorname{sgn}(x) = -1$ if $x < 0$, and otherwise $\operatorname{sgn}(x) = 0$. Compare the bit error rate of the matched filter and the maximum likelihood detectors. Note that the matched filter is also called a single user detector since it can detect the bits of user i without the knowledge of the code vectors of the other users.

11.4.7◆◆ For the CDMA system in Problem 11.3.8, we wish to use MATLAB to evaluate the bit error rate (BER) performance of the decorrelater introduced Problem 11.3.9. In particular, we want to estimate P_e, the probability that for a set of randomly chosen code vectors, that a randomly chosen user's bit is decoded incorrectly at the receiver.

(a) For a k user system with a fixed set of code vectors $\mathbf{S}_1, \ldots, \mathbf{S}_k$, let $\mathbf{S}$ denote the matrix with $\mathbf{S}_i$ as its ith column. Assuming that the matrix inverse $(\mathbf{S}'\mathbf{S})^{-1}$ exists, write an expression for $P_{e,i}(\mathbf{S})$, the probability of error for the transmitted bit of user i, in terms of $\mathbf{S}$ and the $Q(\cdot)$ function. For the same fixed set of code vectors $\mathbf{S}$, write an expression for P_e, the probability of error for the bit of a randomly chosen user.

(b) In the event that $(\mathbf{S}'\mathbf{S})^{-1}$ does not exist, we assume the decorrelator flips a coin to guess the transmitted bit of each user. What are $P_{e,i}$ and P_e in this case?

(c) For a CDMA system with processing gain $n = 32$ and k users, each with SNR 6 dB, write a MATLAB program that averages over randomly chosen matrices $\mathbf{S}$ to estimate P_e for the decorrelator. Note that unlike the case for Problem 11.4.6, simulating the transmission of bits is not necessary. Graph your estimate $\hat{P}_e$ as a function of k.

11.4.8◆◆ Simulate the multi-level QAM system of Problem 11.3.4. Estimate the probability of symbol error and the probability of bit error as a function of the noise variance σ^2.

11.4.9◆◆ In Problem 11.4.5, we used simulation to estimate the probability of *symbol* error. For transmitting a binary bit stream over an MPSK system, we set each $M = 2^N$ and each transmitted symbol corresponds to N bits. For example, for $M = 16$, we map each four-bit input $b_3b_2b_1b_0$ to one of 16 symbols. A simple way to do this is binary index mapping: transmit $\mathbf{s}_i$ when $b_3b_2b_1b_0$ is the binary representation of i. For example, the bit input 1100 is mapped to the transmitted signal $\mathbf{s}_{12}$. Symbol errors in the communication system cause bit errors. For example if $\mathbf{s}_1$ is sent but noise causes $\mathbf{s}_2$ to be decoded, the input bit sequence $b_3b_2b_1b_0 = 0001$ is decoded as $\hat{b}_3\hat{b}_2\hat{b}_1\hat{b}_0 = 0010$, resulting in 2 correct bits and 2 bit errors. In this problem, we use MATLAB to investigate how the mapping of bits to symbols affects the probability of bit error. For our preliminary investigation, it will be sufficient to map the three bits $b_2b_1b_0$ to the $M = 8$ PSK system of Problem 11.3.5.

(a) Find the acceptance sets $\{A_0, \ldots, A_7\}$.

(b) Simulate m trials of the transmission of symbol $\mathbf{s}_0$. Estimate the probabilities $\{P_{0j} | j = 0, 1, \ldots, 7\}$, that the receiver output is $\mathbf{s}_j$ when $\mathbf{s}_0$ was sent. By symmetry, use the set $\{P_{0j}\}$ to determine P_{ij} for all i and j.

(c) Let $\mathbf{b}(i) = \begin{bmatrix} b_2(i) & b_1(i) & b_0(i) \end{bmatrix}$ denote the input bit sequence that is mapped to $\mathbf{s}_i$. Let d_{ij} denote the number of bit positions in which $\mathbf{b}(i)$ and $\mathbf{b}(j)$ differ. For a given mapping, the bit error rate (BER) is

$$\text{BER} = \frac{1}{M} \sum_i \sum_j P_{ij} d_{ij}.$$

(d) Estimate the BER for the binary index mapping.

(e) The Gray code is perhaps the most commonly used mapping:

$\mathbf{b}$	000	001	010	011	100	101	110	111
$\mathbf{s}_i$	$\mathbf{s}_0$	$\mathbf{s}_1$	$\mathbf{s}_3$	$\mathbf{s}_2$	$\mathbf{s}_7$	$\mathbf{s}_6$	$\mathbf{s}_4$	$\mathbf{s}_5$

Does the Gray code reduce the BER compared to the binary index mapping?

11.4.10◆◆ Continuing Problem 11.4.9, in the mapping of the bit sequence $b_2b_1b_0$ to the symbols $\mathbf{s}_i$, we wish to determine the probability of error for each input bit b_i. Let q_i denote the probability that bit b_i is decoded in error. Determine q_0, q_1, and q_2 for both the binary index mapping as well as the Gray code mapping.